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Sugi clear-cut + thinning residue + kiri + CHP — system summary

Working model: `kiri_sugi_integrated_system_model_2026-11-03.xlsx` (v1.2). **All prices and costs are placeholders, not quotes.**

Base case: owners get 30% of sawlog net, 75% of cut-and-left thinning wood collected

- Clear-cut **50 ha/yr** of sugi (600 m³/ha = 30,000 m³, one 931XC at its minimum). Sawlogs (A, B) are sold; C+D wood is chipped for the CHP.
- Half of the cleared land is planted to **kiri** at ¥1M/ha (Tokushima growth, 8.6 t dry/ha/yr, 5-year rotation, coppice replant every 4th cut); the other half to broadleaf (¥0.7M/ha).
- **75% of the thinning wood now left in the forest** (about 6,050 m³/yr) is also chipped, at ¥4,000/t fresh (felling already paid by the village).
- CHP: 25% net electrical efficiency, ¥32/kWh FIT, 50% grant (unconfirmed), sized to the fuel.

Result (¥M/yr)	Years 1–5	Steady state (year 6+)
CHP size	0.58 MWe	**0.75 MWe**
Paid to landowners	34	34
System net	**+48**	**+45**

Cases (steady-state system net, ¥M/yr)

Case	Net	Case	Net
Base	**+45**	Replant kiri every cut	+26
Owners 0% / 50%	+79 / +22	Thinning-type grade split (22% A)	+4
No residue / 50% / 100% residue	+32 / +41 / +49	Low sawlog price, high cost	−10
Kiri at ¥2M/ha	+39	**317 m³/ha, 95 ha/yr** (30,000 m³)	**+1**
		317 m³/ha, 50 ha/yr (15,900 m³)	+6

What drives the result

1. **Volume per hectare.** The model uses 600 m³/ha; the village thinning plan shows only 277–317 m³/ha standing (50–60 yr stands). At 317 m³/ha the base case falls to about break-even.
2. **Grade split.** Sawlog sales fund everything. The 61% A share is the national all-harvest average; Mitsue's neglected stands may be far lower (thinning-type 22% A gives +4).
3. **Landowner share.** Each 10 points costs about ¥11M/yr. 30% is the planning choice, to give owners an incentive.
4. **Residue price and recovery.** The ¥4,000/t placeholder is untested. At ¥8,000/t the residue gain is only about ¥1.6M/yr. Slope and access (winch-assisted 931/855 may help) are unconfirmed.

Not included

Carbon credits, GPU/data-center income, the ¥3,000/t village biomass subsidy, the 931XC/forwarder purchase and crew wages, deer-fence cost beyond the ¥1M/ha planting figure.

What will settle the open inputs

- **Oba (11/6):** forest register / stand volume (B8), stumpage and who receives sale income (B9), clear-cut and replanting rules (B10), A/B log buyers (B11), road access (B5).
- **Komatsu:** slope limits with and without winch, road width, annual output per 931XC + 855 set.
- **Maker RFQ:** real CHP efficiency and capex at 0.6–0.8 MWe (over 1,000 kW would be FIP-only, no splitting).